Input Specification

An input is one mapping from key names to values. A *cell* is a pair (i, ℓ) of a train and a component. The column *Applies to* uses the abbreviations G (Gaussian), IG (inverse Gaussian), W (Wiener), Ga (Gamma), RF (rainflow) and, for a rainflow cell, Mk (Markov), Ca (Cantelli), Ho (Hoeffding), Be (Bernstein), Ch (Chernoff); *all* is every cell. A *cell value* is a scalar, a list of L numbers (one per component, copied to every train) or an $F \times L$ array; a scalar is copied to every cell. A *selector* has the same shapes with strings. A *profile* is defined in Section 1.7; a *flag* is `true` or `false`. No other broadcasting is accepted.

At a cell where a key does not apply, an array entry may be `null` and the key may be omitted; where it applies, `null` is an error. A broadcast value (a scalar, a list, a profile without the axis F) is read only where the key applies. A non-`null` entry of an array with one entry per cell ($F \times L$ array, or profile with the axes F and L) at a cell where the key does not apply is ignored with a `UserWarning`; a selector entry there is still checked (Section 1.12). Key names are case sensitive; a key outside the tables raises an error (Section 1.11).

Sizes and Horizon

Key	Shape	Default	Applies to	Meaning
`F`	integer	required	all	Number of trains; $F > M$
`M`	integer	required	all	Number of missions
`L`	integer	inferred	all	Components per train; the length of the component axis of `model`. Required if `model` is one string and $L > 1$
`H`	integer or `[H1, H2]`	required	all	Horizon; a scalar H means $H_1 = H_2 = H$

The horizon has $T = H_1 + H_2$ days: days $1, \ldots, H_1$ form the transitory phase, days $H_1 + 1, \ldots, T$ the operating phase. A profile has H_2 entries for the operating phase and, optionally, H_1 entries for the transitory phase.

Selectors

Key	Shape	Default	Applies to	Meaning
`model`	selector	required	all	`gaussian`, `inverse_gaussian`, `wiener`, `gamma` or `rainflow`
`repair_model`	selector	`ardinf`	all	Repair law (Section **??**): `ardinf` or `ard1`; every model accepts both
`bound_method`	selector	`cantelli`	RF	Tail bound (Section **??**): `markov`, `cantelli`, `hoeffding`, `bernstein` or `chernoff`
`reliability_impl`	selector	`exact`	G, W, RF (Ca, Be, Ho)	Form of the reliability row (Section **??**): `exact`, `tangent` or `pwl`. A cell with a linear row (IG, Ga, Mk, Ch) is built in the form `exact`; the value is ignored

Key	Shape	Default	Applies to	Meaning
`encoding`	selector	`bigm`	all	Encoding of the dynamics rows (Section **??**): `bigm` or `indicator`
`component_names`	list of L strings	`Component` n for component n	all	Name of each component, used by the plots

Reliability

Key	Shape	Default	Applies to	Meaning
`tau`	cell value	required	all	Degradation threshold $\tau_{i\ell}$
`epsilon`	cell value	required	all	Reliability level $\varepsilon_{i\ell}$: bound on $\Pr(D > \tau_{i\ell})$ on every day
`tangent_ref`	cell value	0.5	cells with form `tangent`	Tangent point $\mu_r = \texttt{tangent_ref} \cdot \tau$
`pwl_points`	integer	8	cells with form `pwl`	Number P of segments

Repair and Starting Anchors

Key	Shape	Default	Applies to	Meaning
`rho`	cell value	required	all	Repair efficiency $\rho_{i\ell}$
`anchor_mu_0`	cell value	μ_0	`ard1` cells	Starting anchor $g^{\mu}_{i\ell 0}$ of the mean
`anchor_v_0`	cell value	v_0	`ard1` cells at G, RF (Ca, Be)	Starting anchor $g^{v}_{i\ell 0}$ of the variance
`anchor_R_0`	cell value	R_0	`ard1` cells at RF (Ho)	Starting anchor $g^{R}_{i\ell 0}$ of the Hoeffding budget
`anchor_K_0`	cell value	K_0	`ard1` cells at RF (Ch)	Starting anchor $g^{K}_{i\ell 0}$ of the Chernoff quantity

Initial State

Key	Shape	Default	Applies to	Meaning
`mu_0`	cell value	required	all	Initial mean degradation $\mu_{0,i\ell}$
`v_0`	cell value	0	G, W, RF (Ca, Be)	Initial variance $v_{0,i\ell}$
`support_0`	cell value	0	RF (Be)	Width $b^0_{i\ell}$ of the range of the initial damage
`R_0`	cell value	0	RF (Ho)	Initial Hoeffding budget $R_{0,i\ell}$: the square of the width of the range of the initial damage
`cgf_0`	cell value	$s\,\mu_0$	RF (Ch)	Initial Chernoff quantity $K_{0,i\ell}$: the cumulant generating function of the initial damage at $s_{i\ell}$

Replacement

Key	Shape	Default	Applies to	Meaning
`replacement_mu`	cell value	0	all	Mean $\mu_{i\ell}^{\text{new}}$ after a replacement
`replacement_v`	cell value	0	G, W, RF (Ca, Be)	Variance $v_{i\ell}^{\text{new}}$ after a replacement
`support_new`	cell value	0	RF (Be)	Width $b_{i\ell}^{\text{new}}$ of the range of the damage after a replacement
`R_new`	cell value	0	RF (Ho)	Hoeffding budget $R_{i\ell}^{\text{new}}$ after a replacement
`cgf_new`	cell value	$s\,\mu^{\text{new}}$	RF (Ch)	Chernoff quantity $K_{i\ell}^{\text{new}}$ after a replacement
`replacement_as_new`	flag	`false`	all	If `true`, every replacement input above must be 0 (the replaced part is in the zero state)
`allow_replacement`	flag	`true`	all	If `false`, the variables x^r and their rows are not built

Increment Profiles and the Phase Lookup

A profile gives a parameter of the damage increment when train i flies mission $j \in \{1, \dots, M\}$, for component ℓ. Its canonical shape is (F, L, M, H_*), with $H_* = H_2$ for an operating profile and $H_* = H_1$ for a transitory profile. The parser accepts the shapes scalar, (M), (L, M), (L, M, H_*), (F, L, M) and (F, L, M, H_*), and expands each of them to (F, L, M, H_*) before any row is built (eager broadcasting); a missing axis is repeated. An array whose shape fits both (L, M, H_*) and (F, L, M) is read as (L, M, H_*). `gamma_beta` and `gamma_beta_trans` read a list as one rate per component and a two-dimensional array as one rate per cell; rates per mission have the shape (F, L, M).

Let p be an operating profile and p^{tr} the transitory profile of the same key, if given. The value used on day $k \in \{1, \dots, T\}$ is

$$p_{i\ell jk}^{\text{inc}} = \begin{cases} p_{i\ell jk}^{\text{tr}} & k \le H_1 \text{ and } p^{\text{tr}} \text{ is given,} \\ p_{i\ell j,\,((k-1) \bmod H_2)+1} & k \le H_1 \text{ and } p^{\text{tr}} \text{ is not given,} \\ p_{i\ell j,\,((k-H_1-1) \bmod H_2)+1} & k > H_1. \end{cases} \tag{1}$$

Each profile key is looked up on its own. The symbols $\mu_{i\ell jk}^{\text{inc}}$, $v_{i\ell jk}^{\text{inc}}$, $b_{i\ell jk}^{\text{inc}}$ and $K_{i\ell jk}^{\text{inc}}$ denote the values after the lookup.

Key	Shape	Default	Applies to	Meaning
`mu`	profile	required	all	Mean $\mu_{i\ell jk}^{\text{inc}}$ of the increment; for Ga the increment has shape $\mu\,\beta$ and rate β
`v`	profile	required	G, RF (Ca, Be)	Variance $v_{i\ell jk}^{\text{inc}}$ of the increment
`support`	profile	required	RF (Ho, Be)	Width $b_{i\ell jk}^{\text{inc}}$ of the range of the increment
`cgf`	profile	required	RF (Ch)	Cumulant generating function $K_{i\ell jk}^{\text{inc}} = \ln \mathbb{E}\, e^{sX}$ of the increment X at $s = s_{i\ell}$

Key	Shape	Default	Applies to	Meaning
`s_chernoff`	cell value	required	RF (Ch)	Chernoff parameter $s_{i\ell}$
`gamma_beta`	scalar, (L), (F, L), (F, L, M) or (F, L, M, H_2)	required	Ga	Rate $\beta_{i\ell jk}$ of the Gamma increment; a shape without a day axis is repeated along the days
`mu_trans`, `v_trans`, `support_trans`, `cgf_trans`	transitory profile	none	as the key without the suffix	Transitory-phase profile of the key; it requires the operating profile
`gamma_beta_trans`	scalar, (L), (F, L), (F, L, M) or (F, L, M, H_1)	none	Ga	Rates of the transitory phase

Model Constants

Key	Shape	Default	Applies to	Meaning
`sigma`	cell value	required	W	Diffusion constant $\sigma_{i\ell}$: every mission adds $\sigma_{i\ell}^2$ to the variance
`eta`	cell value	required	IG	Dispersion $\eta_{i\ell}$ of the inverse Gaussian law of each term (Section **??**)
`v_max_user`	cell value	none	G, W, RF (Ca, Be)	Upper bound $V^{\text{user}}_{\max,i\ell}$ on the variance; if given, the row $v_{i\ell k} \le V^{\text{user}}_{\max,i\ell}$ holds on every day
`gamma_beta_0`	cell value	none (known damage)	Ga	Rate of the initial damage, a Gamma variable with mean μ_0; if omitted, the initial damage is the known constant μ_0 and does not enter the smallest rate
`gamma_beta_new`	cell value	none (known damage)	Ga	Rate of the damage after a replacement, a Gamma variable with mean μ^{new}; if omitted, it is the known constant μ^{new} and does not enter the smallest rate

Costs and Objective

Key	Shape	Default	Applies to	Meaning
`C_M`	scalar	required	all	Cost of one depot day of one train, whatever the number of jobs on that day
`C_R`	cell value	required	all	Repair cost $C_{R,i\ell}$ per unit of degradation removed; at a cell with $C_{R,i\ell} = 0$ the variable z and its rows are not built
`C_rep`	cell value	required	all	Replacement cost $C_{\text{rep},i\ell}$

Key	Shape	Default	Applies to	Meaning
`C_D`	scalar	required	all	Damage regularisation coefficient C_D; if $C_D = 0$, the variable u and its rows are not built and `penalty_type` is ignored
`penalty_type`	string	`inf_norm`	all	Damage regularisation (Section **??**): `inf_norm` or `quadratic`; if $C_D > 0$, `quadratic` requires `reliability_impl exact` at every cell with a quadratic reliability row (G, W, RF with Ca, Ho or Be); another value is an error

Run Options

Key	Shape	Default	Meaning
`verbose`	integer	1	Gurobi output flag: 0 or 1
`mip_gap`	number	0.1	Relative MIP gap at which Gurobi stops
`time_limit`	number	3600	Time limit of Gurobi, in seconds
`fast`	flag	`false`	If `true`: `MIPFocus` = 1, `Heuristics` = 0.5, `ImproveStartGap` = 0.5, `NoRelHeurTime` = max(2, 0.15 `time_limit`)
`gurobi_params`	mapping	none	Gurobi parameters by name, set after every other option
`objective_mode`	string	`total`	`total`, `operating_average` or `evaluation_total` (Section **??**)
`evaluation_horizon`	integer	none	Evaluation horizon E; required if `objective_mode` is `evaluation_total`
`progress_interval_seconds`	number	5.0	Minimum time between two records of the progress report (Section 2.3)
`relaxation_warm_start`	flag	`false`	If `true` and no other warm start is applied, a relaxation start is computed (Section **??**)
`relaxation_time_limit`	number	60.0	Time limit of the relaxation start, in seconds
`validation`	mapping	none	Settings of the Gamma validators (Section **??**): `mode`, `repetitions`, `random_seed`, `maximum_schedule_failure_rate`, `confidence_level`, `batch_size`; `solve` does not read it

No input key selects a warm start from a previous result (Section **??**).

Rejected Keys

A key of the table raises an error whose message names the canonical key and, where the meaning differs, the meaning of the canonical key. Any other key outside the tables raises the error "unknown key" with the list of the canonical keys. The check on keys runs before any array is read.

Rejected key	Message: canonical key and meaning
`maintenance_type`	Use `repair_model`. Its values are `ardinf` (the repair reduces the whole level) and `ard1` (it reduces the level accumulated since the last intervention). In a file written with `maintenance_type`, the value `ARD1` corresponds to `ardinf`, and the value `ARA1` to `ard1`
`tail_bound, method`	Use `bound_method`; its values are `markov`, `cantelli`, `hoeffding`, `bernstein` and `chernoff`
`formulation`	Use `reliability_impl`. The value `exact` keeps its meaning. The value `lp` corresponds to `tangent` with `tangent_ref` equal to 0
`beta`	Use `gamma_beta`. It is a rate, and `beta` is a scale: `gamma_beta` equals 1/`beta`
`alpha_inc`	The Gamma increments are given by the mean `mu` and the rate `gamma_beta`. The shape of an increment is `mu` times `gamma_beta`
`gamma_beta_bound`	Remove it. The formulation computes the smallest rate of each cell from the rates of the cell (Section **??**)
`gamma_calibration_method`	Remove it. The formulation has no calibration step (Section **??**)
`b_inc`	Use `support`: the width of the range of the increment
`b_0`	Use `support_0`: the width of the range of the initial damage
`b_new`	Use `support_new`: the width of the range of the damage after a replacement
`n_workers, warm_start`	Remove them. Parallel runs and warm starts between horizons are options of the sweep functions (Sections **??** and **??**)
`depot_capacity`	Remove it. The availability of the depot is not limited at fleet level (Section **??**)
`C_P`	Remove it. The loop constraint is a hard constraint and has no penalty (Section **??**)
`C_S`	Use `C_D`, with the same meaning
`alpha`	Use `tau`, with the same meaning
`xi, repair_rho`	Use `rho`, with the same meaning
`mu_new`	Use `replacement_mu`, with the same meaning
`v_new`	Use `replacement_v`, with the same meaning

Consistency Checks

The solver validates all of the following before it builds the model. A violation raises a `ValueError` whose message names the key and, for a cell value, the cell (i, ℓ). A check of one model runs only at the cells of that model, on the entries where the key applies.

Keys, shapes and entries.

- Every key is in the tables of Section 1. The keys `F`, `M`, `H`, `model`, `tau`, `epsilon`, `rho`, `mu_0`, `mu`, `C_M`, `C_R`, `C_rep`, `C_D` and the keys that each cell type requires (Section 3) are present.
- Each value has one of the shapes of its table row; the last axis of an operating profile has H_2 entries and that of a transitory profile H_1; `component_names` has L entries.
- At a cell where a key applies, every entry is a finite number (a string for a selector); `null` is an error.

Sizes and selectors.

- F, M and L are positive integers, and $F > M$. A given `L` equals the length of the component axis of `model` when `model` is a list or an array.
- Every entry of `component_names` is a string that is not empty after blanks are removed.
- H is an integer ≥ 1, or a pair of integers $H_1, H_2 \geq 1$.

- `model` entries are `gaussian`, `inverse_gaussian`, `wiener`, `gamma` or `rainflow`; `repair_model` entries are `ard1` or `ardinf`, for every model.
- `bound_method` entries at rainflow cells are `markov`, `cantelli`, `hoeffding`, `bernstein` or `chernoff`.
- Every non-`null` entry of `reliability_impl` is `exact`, `tangent` or `pwl`, also at a cell whose row is linear (IG, Ga, Mk, Ch); there the value is ignored and the form `exact` is built. `encoding` entries are `bigm` or `indicator`.
- `penalty_type` is `inf_norm` or `quadratic`. If $C_D > 0$ and `penalty_type` is `quadratic`, `reliability_impl` is `exact` at every cell with a quadratic reliability row (G, W, and RF with Ca, Ho or Be); another value is an error.

Reliability, repair and initial state.

- $\tau_{i\ell} > 0$ and finite.
- $\varepsilon_{i\ell} \in (0, 1)$, and $\varepsilon_{i\ell} < 0.5$ at G and W cells.
- $\rho_{i\ell} \in (0, 1]$.
- $0 \le \mu_{0,i\ell} < \tau_{i\ell}$.
- v_0, `support_0` and R_0 are ≥ 0; if `allow_replacement` is `true`, so are `replacement_mu`, `replacement_v`, `support_new` and R^{new}.
- `tangent_ref` $\in [0, 1)$ at cells with the form `tangent`; `pwl_points` is an integer ≥ 1.
- Starting anchors: $0 \le$ `anchor_mu_0` $\le \mu_0$, $0 \le$ `anchor_v_0` $\le v_0$, $0 \le$ `anchor_R_0` $\le R_0$ and $0 \le$ `anchor_K_0` $\le K_0$, at the cells that read them.
- If `replacement_as_new` is `true`, `replacement_mu`, `replacement_v`, `support_new`, `R_new` and `cgf_new` are 0 at every cell that reads them; the check does not run when the flag is `false`.

Natural ranges and replacement states.

- Every initial value (μ_0, v_0, R_0, K_0) and every replacement value (μ^{new}, v^{new}, R^{new}, K^{new}; only if `allow_replacement` is `true`) lies in $[0, M_s]$, with the natural range M_s of its state: τ for a mean; $V_{\max} = \tau^2/\big(\Phi^{-1}(1-\varepsilon)\big)^2$ for the variance at G and W cells; $\varepsilon\tau^2/(1-\varepsilon)$ at Ca cells; $\tau(\tau - 2c)/(2\kappa)$ at Be cells, with $c = \kappa b/3$ and $\kappa = \ln(1/\varepsilon)$; $2\tau^2/\kappa$ for R at Ho cells; $s\tau - \kappa$ for K at Ch cells (Sections **??** to **??**). Hence every starting anchor lies in its range.
- If `allow_replacement` is `true`, the replacement state $(\mu^{\text{new}}, v^{\text{new}}, R^{\text{new}}, K^{\text{new}})$ of every cell satisfies every row that holds on a replacement day, in the form built at the cell: the reliability row in the form of `reliability_impl` (for `pwl`, the cap of the segment that contains μ^{new}), its side row, and $v^{\text{new}} \le V^{\text{user}}_{\max,i\ell}$ where `v_max_user` is given (Sections **??** to **??** and **??**).

Increments.

- $0 \le \mu^{\text{inc}}_{i\ell jk} < \tau_{i\ell}$ for every mission j and every day $k \in \{1, \dots, T\}$.
- Where $\mu^{\text{inc}}_{i\ell jk} = 0$, the values $v^{\text{inc}}_{i\ell jk}$, $b^{\text{inc}}_{i\ell jk}$ and $K^{\text{inc}}_{i\ell jk}$ are 0.
- $v^{\text{inc}}_{i\ell jk}$, $b^{\text{inc}}_{i\ell jk}$ and $K^{\text{inc}}_{i\ell jk}$ are ≥ 0. A transitory profile has its operating profile.

Gaussian, inverse Gaussian, Wiener and Gamma cells.

- G: $\mu_0 \ge 3\sqrt{v_0}$, $\mu^{\text{inc}}_{i\ell jk} \ge 3\sqrt{v^{\text{inc}}_{i\ell jk}}$ for every mission j and day k, and, if `allow_replacement` is `true`, $\mu^{\text{new}} \ge 3\sqrt{v^{\text{new}}}$.
- IG: $\eta_{i\ell} > 0$.

- W: $\sigma_{i\ell} > 0$ and $v_{0,i\ell} \geq 0$.
- Ga: every entry of `gamma_beta` and `gamma_beta_trans` is finite and > 0; `gamma_beta_0` and `gamma_beta_new`, where given, are finite and > 0.
- G, W, RF (Ca, Be) with `v_max_user` given: $V^{\text{user}}_{\max,i\ell} > 0$, $V^{\text{user}}_{\max,i\ell} \geq v_{0,i\ell}$ and, if `allow_replacement` is `true`, $V^{\text{user}}_{\max,i\ell} \geq v^{\text{new}}_{i\ell}$.

Rainflow cells.

- Be: $v^{\text{inc}}_{i\ell jk} \leq (b^{\text{inc}}_{i\ell jk})^2/4$ for every mission j and day k, $v_0 \leq (b^0)^2/4$ and, if `allow_replacement` is `true`, $v^{\text{new}} \leq (b^{\text{new}})^2/4$.
- Be: $\tau_{i\ell} > 2\kappa\, b_{i\ell}/3$, with $\kappa = \ln(1/\varepsilon_{i\ell})$ and $b_{i\ell}$ the support constant of Section **??**.
- Be with the form `tangent`: `tangent_ref` $\cdot\, \tau_{i\ell} < \sqrt{\tau_{i\ell}\,(\tau_{i\ell} - 2c_{i\ell})}$, with $c_{i\ell} = \kappa\, b_{i\ell}/3$; Be with the form `pwl`: $\tau_{i\ell}/(2P) < \sqrt{\tau_{i\ell}\,(\tau_{i\ell} - 2c_{i\ell})}$, with $P =$ `pwl_points`.
- Ch: `s_chernoff` is given and $s_{i\ell} > \ln(1/\varepsilon_{i\ell})/\tau_{i\ell}$.
- Ch: $K^{\text{inc}}_{i\ell jk} \geq s_{i\ell}\, \mu^{\text{inc}}_{i\ell jk}$ for every mission j and day k, $K_0 \geq s\,\mu_0$, at `ard1` cells $g^K_0 \geq s\, g^{\mu}_0$, and, if `allow_replacement` is `true`, $K^{\text{new}} \geq s\,\mu^{\text{new}}$.

Run options and costs.

- `objective_mode` is `total`, `operating_average` or `evaluation_total`; in the last mode `evaluation_horizon` is given. If `evaluation_horizon` is given, in any mode, it is an integer $E > H_1$.
- `mip_gap` ≥ 0; `time_limit`, `progress_interval_seconds` and `relaxation_time_limit` are > 0; `gurobi_params` and `validation` are mappings; `fast`, `relaxation_warm_start`, `allow_replacement` and `replacement_as_new` are flags.
- C_M, C_D, $C_{R,i\ell}$ and $C_{\text{rep},i\ell}$ are finite and ≥ 0.

Output Specification

The function `solve()` returns one result dictionary for the horizon H of the input, and writes it to a file if `results_path` is not `None`. Array values are `numpy.ndarray` objects; the result holds no Gurobi object. In the column *Shape*, a dash marks a scalar or a string.

Result Dictionary

Key	Shape	Description
Status and quality		
`status`	—	`optimal`, `time_limit`, `suboptimal`, `interrupted`, `solution_limit`, `node_limit`, `iteration_limit`, `infeasible`, `inf_or_unbounded`, `unbounded`, or `gurobi_status_`n for any other Gurobi status code n
`objective`, `mip_gap`, `bound`	— or `null`	Objective value of the best solution found; relative MIP gap reached; best bound, if available
Problem		
`F`, `M`, `L`, `H1`, `H2`, `T`	—	F, M, L, H_1, H_2 and $T = H_1 + H_2$
`H`	integer or $[H_1, H_2]$	Horizon as written in the input
`model`	$F \times L$ list of strings	Degradation model of each cell, as in the input
`component_names`	list of L strings	Names of the components

Key	Shape	Description
`tau`, `mu_0`, `v_0`	$F \times L$	Threshold, initial mean and initial variance of each cell, after broadcasting
`bound_method`, `repair_model`, `reliability_impl`, `encoding`	string, or $F \times L$ list of strings	Selector of each cell after broadcasting: one string if all cells agree, else a list; `bound_method` is `null` at a cell that is not rainflow
`objective_mode`, `evaluation_horizon`	—	Objective mode, and E if given
Solution (`null` if no solution was found)		
`x`	$F \times (M{+}1) \times T$	Assignment x_{ijk} (Section **??**)
`m`	$F \times L \times T$	Repair $x^m_{i\ell k}$
`r`	$F \times L \times T$	Replacement $x^r_{i\ell k}$; zero if `allow_replacement` is `false`
`idle`	$F \times L \times T$	$\text{idle}_{i\ell k} = x_{i0k} - x^m_{i\ell k} - x^r_{i\ell k}$: component ℓ is in the depot without an intervention
`mu`	$F \times L \times T$	Tracked mean degradation
`v`	$F \times L \times T$	Tracked variance; `NaN` at cells that track no variance (IG, Ga, RF with Mk, Ho or Ch)
`z`	$F \times L \times T$	Repair-cost variable; `NaN` at cells with $C_{R,i\ell} = 0$. On the transitory days of the mode `operating_average`: Equation (2)
`R`, `K`	$F \times L \times T$	Tracked Hoeffding budget (cells RF (Ho)) and Chernoff quantity (cells RF (Ch)); `NaN` elsewhere
`g_mu`, `g_v`, `g_R`, `g_K`	$F \times L \times T$	Tracked anchors $g^\mu_{i\ell k}$, $g^v_{i\ell k}$, $g^R_{i\ell k}$, $g^K_{i\ell k}$ (Section **??**), only at `ard1` cells: `g_mu` at every model, `g_v` at G and RF (Ca, Be), `g_R` at RF (Ho), `g_K` at RF (Ch); `NaN` elsewhere
`u`	—	Damage regularisation value, Equation (3)
Cost report (`null` if no solution was found)		
`step_costs`	T	Cost c_k of each day, Equation (**??**)
`J_initialization`, `J_op`, `J_op_average`	—	J_{init}, J_{op} and J_{op}/H_2, Equation (**??**)
`peak_damage`, `damage_penalty`	—	u and $C_D\,u$
`operating_objective`, `J_total`	—	$J_{\text{op}}/H_2 + C_D\,u$ and $J_{\text{init}} + J_{\text{op}} + C_D\,u$
`projected_evaluation_cost`, `evaluation_objective`	— or `null`	P_E (Equation (**??**)) and $P_E + C_D\,u$; `null` if `evaluation_horizon` is not given
`component_costs`	mapping	Cost coefficients used: `C_M`, `C_D` (scalars), `C_R`, `C_rep` ($F \times L$)
Reports (always present)		
`performance`	mapping	Sizes and times of the solve
`optimization_progress`	list of mappings	Progress of the incumbent and the bound
`warm_start`, `relaxation_warm_start`	mappings	What each warm start did

Rules for the Values

If Gurobi found no solution, the solution arrays and the cost report are `null`, and so are `objective` and `mip_gap`; if it found one, the arrays hold the best solution and `status` tells how

the solve ended. The arrays `mu`, `v`, `R`, `K`, `g_mu`, `g_v`, `g_R`, `g_K` and `z` hold the tracked variables of the model, and are upper bounds (Section **??**).

In the mode `operating_average`, z is not built on the days of the transitory phase (Section **??**); at a cell with $C_{R,i\ell} > 0$ the result holds, for those days, the value computed after the solve:

$$z_{i\ell k} = \begin{cases} \rho_{i\ell}\left(\mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1}\right) & \text{if } x^{m}_{i\ell k} = 1, \\ 0 & \text{otherwise,} \end{cases} \qquad k = 1, \ldots, H_1, \tag{2}$$

with $\mu_{i\ell 0} = \mu_{0,i\ell}$, $g^{\mu}_{i\ell 0}$ the starting anchor, and $g^{\mu} \equiv 0$ under `ardinf`.

The value u is recomputed from `mu`, and is reported also when $C_D = 0$:

$$u = \max_{i,k} \sum_{\ell=1}^{L} \mu_{i\ell k} \quad (\texttt{inf_norm}), \qquad u = \max_{i,k} \sum_{\ell=1}^{L} \mu^{2}_{i\ell k} \quad (\texttt{quadratic}). \tag{3}$$

The cost c_k of day k and the phase costs J_{init}, J_{op} are Equations (**??**) and (**??**) (Section **??**) on the arrays of the result, with $z_{i\ell k} = 0$ where it is `NaN`. When the variable u has no slack, `objective` equals, up to the tolerance of the solver, `J_total`, `operating_objective` or `evaluation_objective` for the mode `total`, `operating_average` or `evaluation_total`; otherwise it is larger by C_D times the slack.

Reports

`performance` is a mapping; an entry that Gurobi does not give is `null`. *Keys:*

- size entries: `gurobi_version`; `variables`, with `continuous_variables`, `integer_variables` and `binary_variables`; the rows `linear_constraints`, `general_constraints`, `indicator_constraints` and `quadratic_constraints`; `nonzeros`;
- work entries: `solutions_found` and `gurobi_runtime_seconds`; the counts `branch_and_bound_nodes`, `simplex_iterations` and `barrier_iterations`; `work_units`;
- wall-clock entries, in seconds: `model_construction_seconds`, `optimizer_call_seconds`, `solution_extraction_seconds` and `backend_wall_seconds`;
- settings: `requested_time_limit_seconds`, `requested_mip_gap`, `threads` and `seed`.

`optimization_progress` is a list of records with the entries `runtime_seconds`, `incumbent`, `best_bound`, `relative_gap`, `nodes` and `solutions`; the incumbent, the bound and the gap are `null` while Gurobi has none. A record is made at the first progress report of Gurobi, then at least `progress_interval_seconds` after the previous record and whenever the number of solutions changes; the last record is the final state of the solve.

`warm_start` (Section **??**) is a mapping with the entries `applied` (`true` when at least one value was set), `values` (number of values set), `mode` (`"start"`, `"hint"` or `"both"`, `null` without a source), `source_horizon` (the pair H_1, H_2 of the source, or `null`) and `completion`. `completion` (Section **??**) is a mapping with `requested`, `status`, `fixed_integer_variables`, `values_copied`, `runtime_seconds` and `iis_constraints`. Its `status` is one of `"not_requested"`, `"completed"`, `"infeasible"`, `"infeasible_or_unbounded"` and `"time_limit"`, or `"status_`q`"` for another Gurobi status code q. `iis_constraints` is a list of at most 50 names of rows of an irreducible infeasible subset, empty unless the status is `"infeasible"`.

`relaxation_warm_start` (Section **??**) is a mapping with the entries `enabled` (the value of the option); `status` (`"disabled"` if the option is false, `"skipped"` if it is true and a start or a hint was applied, otherwise the Gurobi status of the relaxed copy as a string); `relaxation_objective`; `values_copied`; `runtime_seconds`.

Result Files

`solve` writes the result to the file `results_path`.

- The default of `results_path` is `"output.yaml"`; `None` writes no file.
- A path without extension gets the extension `.yaml`. `.yaml` and `.yml` give YAML, `.json` gives JSON, `.h5` and `.hdf5` give HDF5; another extension raises a `ValueError`.
- The extension and the directory of the path are checked before the input file is read: the directory must exist (`FileNotFoundError`) and be writable (`PermissionError`).
- YAML and JSON files hold plain data (lists, numbers, strings, `null`); `NaN` is written as `null`. They load with `yaml.safe_load` and `json.load`. HDF5 files keep `NaN` (Section 3.2).
- A key whose value is `None` is not written, except `objective`, written as `null`. Every reader of a result treats a missing key as `null`.

Input File Schema

Format Rules

The formats are YAML (`.yaml`, `.yml`), JSON (`.json`) and HDF5 (`.h5`, `.hdf5`), chosen by the extension of the path in lower case; another extension raises an error. One reader serves every function that takes an input path.

- A file holds one mapping from the keys of Section 1 to values, with the same keys and meaning in the three formats.
- An array is a nested list in YAML and JSON; the outermost list runs over the first axis of the shape (train, component, mission, day).
- YAML is read in the safe mode.

HDF5 Layout

An HDF5 file is flat: every key is an attribute or a dataset at the root. The reader makes one entry of every root attribute and root dataset; a byte string is decoded as UTF-8, and a text attribute that begins with `{` or `[` is parsed as JSON.

Input file.

- A scalar number, a flag or a single string is an attribute, or a dataset of shape (); `H` is one integer or two integers.
- An array of numbers is a `float64` dataset with the shape of the key; a `null` entry is `NaN`.
- A selector array and `component_names` are datasets of strings, or JSON text; a `null` entry of a selector is the empty string. `gurobi_params` and `validation` are JSON text.

Output file. A number or a string is an attribute; an array of numbers is a `float64` dataset, with `NaN` kept; a list of strings, a mapping or a list of records is JSON text in an attribute; a key whose value is absent is not stored.

Placement	Keys
Attribute: number or string	`status`, `objective`, `mip_gap`, `bound`, `F`, `M`, `L`, `H` (one or two integers), `H1`, `H2`, `T`, `objective_mode`, `evaluation_horizon`, `u`, `J_initialization`, `J_op`, `J_op_average`, `peak_damage`, `damage_penalty`, `operating_objective`, `J_total`, `projected_evaluation_cost`, `evaluation_objective`; the selectors `bound_method`, `repair_model`, `reliability_impl`, `encoding` when they are one string
Dataset (`float64`)	`x`, `mu`, `v`, `R`, `K`, `g_mu`, `g_v`, `g_R`, `g_K`, `z`, `m`, `r`, `idle`, `step_costs`, `tau`, `mu_0`, `v_0`
JSON text in an attribute	`model`, `component_names`, `component_costs`, `performance`, `optimization_progress`, `warm_start`, `relaxation_warm_start`; the selectors `bound_method`, `repair_model`, `reliability_impl`, `encoding` when they differ between cells

Keys by Cell Type

The table lists the keys that each cell type requires, and the optional keys that it reads in addition to those of the first row. `*_trans` is optional for every profile key; `tangent_ref` and `pwl_points` are read only at cells with the form `tangent` or `pwl`; the starting anchors are read only at `ard1` cells.

Cell type	Required	Optional (defaults in Section 1)
Every cell	`F`, `M`, `H`, `model`, `tau`, `epsilon`, `rho`, `mu_0`, `mu`, `C_M`, `C_R`, `C_rep`, `C_D`	`L`, `repair_model`, `reliability_impl`, `encoding`, `tangent_ref`, `pwl_points`, `anchor_mu_0`, `replacement_mu`, `replacement_as_new`, `allow_replacement`, `penalty_type`, `component_names`, `mu_trans`, the run options
G	`v`	`v_0`, `replacement_v`, `anchor_v_0`, `v_max_user`, `v_trans`
IG	`eta`	none
W	`sigma`	`v_0`, `replacement_v`, `v_max_user`
Ga	`gamma_beta`	`gamma_beta_0`, `gamma_beta_new`, `gamma_beta_trans`
RF, Mk	`bound_method` with the value `markov`	none
RF, Ca	`v`	`v_0`, `replacement_v`, `anchor_v_0`, `v_max_user`, `v_trans`
RF, Ho	`support`	`R_0`, `R_new`, `anchor_R_0`, `support_trans`
RF, Be	`v`, `support`	`v_0`, `support_0`, `replacement_v`, `support_new`, `anchor_v_0`, `v_max_user`, `v_trans`, `support_trans`
RF, Ch	`cgf`, `s_chernoff`	`cgf_0`, `cgf_new`, `anchor_K_0`, `cgf_trans`

YAML Example (Mixed Models)

Three trains, two missions, two components; the six cells use G, IG, W, Ga, RF with Be and RF with Ch. The example passes every check of Section 1.12.

```
F: 3
M: 2
L: 2
H: [2, 3]                # H1 = 2, H2 = 3, so T = 5
component_names: [Wheelset, Brake pad]
```

```
model: [[gaussian, inverse_gaussian], [wiener, gamma], [rainflow,
   rainflow]]
repair_model: [[ard1, ard1], [ardinf, ard1], [ard1, ard1]]
bound_method: [[null, null], [null, null], [bernstein, chernoff]]
reliability_impl: [[tangent, null], [pwl, null], [exact, null]]
encoding: [[bigm, bigm], [bigm, bigm], [bigm, indicator]]

tau: 1.0
epsilon: [[0.05, 0.05], [0.05, 0.05], [0.10, 0.10]]
tangent_ref: 0.4        # tangent cell only
pwl_points: 6           # pwl cell only
rho: [0.8, 0.7]         # per component, the same for every train

mu_0: [[0.12, 0.10], [0.15, 0.08], [0.09, 0.14]]
v_0: [[0.0004, null], [0.0005, null], [0.0004, null]]
support_0: [[null, null], [null, null], [0.05, null]]
anchor_mu_0: [[0.10, 0.06], [null, 0.05], [0.00, 0.10]]
anchor_v_0: [[0.0002, null], [null, null], [0.0002, null]]
anchor_K_0: [[null, null], [null, null], [null, 0.50]]

replacement_mu: 0.02
replacement_v: [[0.00004, null], [0.00004, null], [0.00004, null]]
support_new: [[null, null], [null, null], [0.03, null]]
replacement_as_new: false
allow_replacement: true

mu: [[0.06, 0.05], [0.04, 0.07]]        # profile (L, M)
mu_trans: [[[0.07, 0.06], [0.06, 0.05]], [[0.04, 0.04], [0.07, 0.07]]]
   # (L, M, H1)
v: [[0.0001, 0.0001], [null, null]]
support: [[0.05, 0.04], [null, null]]
cgf: [[null, null], [0.22, 0.38]]       # at s_chernoff = 5
s_chernoff: [[null, null], [null, null], [null, 5.0]]

sigma: [[null, null], [0.03, null], [null, null]]
eta: [[null, 0.15], [null, null], [null, null]]
v_max_user: [[null, null], [0.05, null], [null, null]]
gamma_beta: [[null, null], [null, 20.0], [null, null]]
gamma_beta_new: [[null, null], [null, 20.0], [null, null]]  # no
   gamma_beta_0

C_M: 1.0
C_R: [2.0, 1.5]
C_rep: [10.0, 8.0]
C_D: 1.5
penalty_type: inf_norm
objective_mode: evaluation_total
evaluation_horizon: 30

verbose: 0
mip_gap: 0.05
time_limit: 600
fast: false
gurobi_params: {Threads: 4}
progress_interval_seconds: 10
relaxation_warm_start: false
```

References

References

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